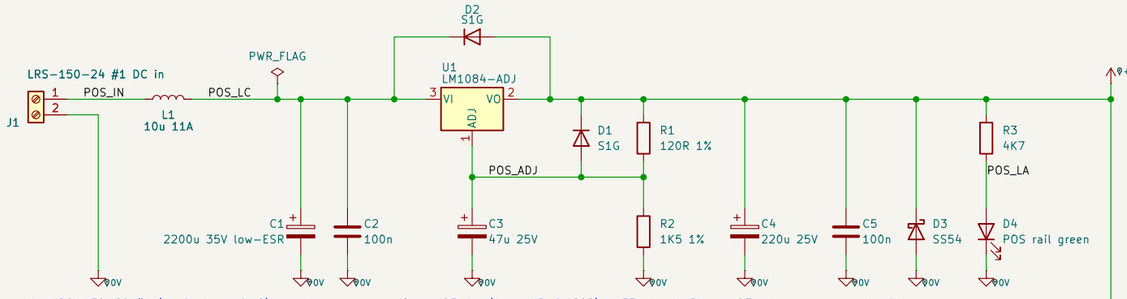
POS RAIL LRS-150-24 #1 (21.6 V) -> LC -> LM1084-ADJ 17.0 V, soft start; OUT = +V, RTN = 0V



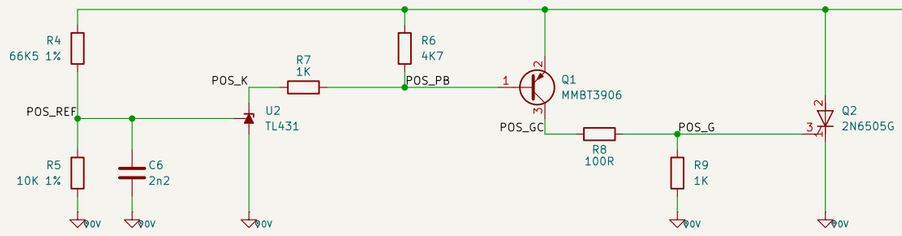
J1: LRS-150-24 #1 (+ pin 1, - pin 2), set to 21.6 V; 150 mV pk-pk ripple at about 65 kHz; 5.1 A peak at start-up (its limit: >= 6.8 A).
L1 / C1: -47 dB, 0.7 mV pk-pk at the LDO input.

VO = 1.25 V x (1 + 1K5 / 120R) + 55 uA x 1K5
= 17.0 V; 120R is also U1's 10 mA minimum load.
C3: soft start: 10 % at 5 ms, 90 % at 165 ms.
D1 / D2 let C3 and C4 discharge around U1.
R2: 1206 0.5 W: 15.7 V across it is 164 mW.

D3: clamps a reversed rail.
D4 lit: rail up. Flashing: the crowbar has fired and the SMPS hiccups into the fault.

+V = 17.0 V @ (, 3.31, 4.54) A (min, typ, max); 4.54 A is the accepted worst corner; design 4 A per rail.

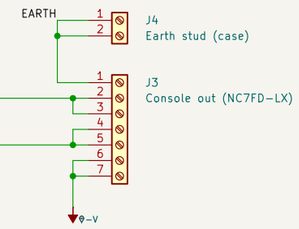
POS CROWBAR TL431 at 66K5 / 10K = 19.1 V -> PNP -> SCR gate; SCR 2N6505G across the rail



Trips at 2.495 V x (1 + 66K5 / 10K) = 19.1 V (18.6-19.9 V worst case): 2.1 V over the output, 0.9 V under the OPA1612's 20 V abs max.
Running: POS_REF at 2.21 V, the gate never moves. A TL431, not a zener: the gate's up to 30 mA would move a zener's trip by over 1 V.

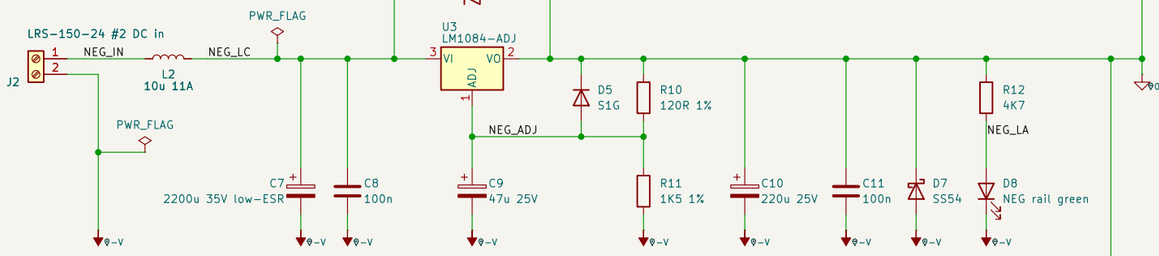
Q2 2N6505G (25 A, 250 A surge) fires 17 us after a shorted pass element (20.7 V peak at the terminals) and holds the rail near 1 V; the SMPS hiccups and D4 flashes: switch off and repair.

OUTPUT 7-way terminal to the NC7FD-LX on the PSU panel: 1 earth, 2-3 +V, 4-5 0 V, 6-7 -V (0 V not earthed here)



J3: to the NC7FD-LX (female) on the PSU panel, each rail on two contacts; cable NC7MX - NC7FX, 7 x 1.0 mm2, <= 2 m.
0V is not earthed in the PSU: it meets mains earth only at the console's star point.
J4: to the earth stud on the case, where the LRS-150 FG terminals go too; J3 pin 1 takes earth to the console.

NEG RAIL LRS-150-24 #2 (21.6 V) -> LC -> LM1084-ADJ 17.0 V, soft start; OUT = 0V, RTN = -V



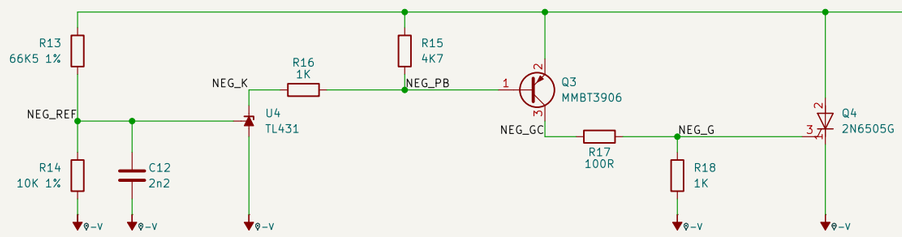
J2: LRS-150-24 #2 (+ pin 1, - pin 2), isolated from #1: its - output is -V, so this copy of the POS rail makes 0V, 17.0 V above -V.

U3 holds 0V at 17.0 V above -V, as U1 holds +V above 0V; U3's tab is at 0V.
R11: 1206 0.5 W, as R2.

D7: clamps a reversed rail.
D8: as D4.

0V: the NEG regulator's output, 17.0 V above -V. -V = -17.0 V @ (, 2.79,) A (min, typ, max); design 4 A.

NEG CROWBAR TL431 at 66K5 / 10K = 19.1 V -> PNP -> SCR gate; SCR 2N6505G across the rail



Trips when 0V rises 19.1 V above -V: the POS crowbar's figures, between 0V and -V.

Q4 fires as Q2 does and holds 0V near 1 V above -V.

Layout: the 4 A paths (J1 / J2 -> L -> LDO -> J3) want >= 3 mm tracks or pours on both layers.
Heatsink: U1, Q2, U3, Q4 (TO-220) share one, each on a silicone pad and a shoulder washer:
the tabs sit at +17 V (U1), 0 V (U3) and the rails (SCR anodes). U1 and U3: 19 W each at 4 A, 12 W typical.
Stacking: the two SMPS outputs are isolated, so both rails are the same positive regulator and crowbar: POS OUT = +V over RTN = 0V, NEG OUT = 0V over RTN = -V.
The SMPS modules, the mains wiring, the NC7 socket and the earth stud are chassis parts.

Two floating rails: LRS-150-24 -> LC -> LM1084-ADJ -> TL431/SCR crowbar
Generated by tools/gen_psu.py from psu_spec.md v0.2 (option B)

Sheet: /
File: psu.kicad_sch

Title: External PSU regulator board - goa console (+-17 V, 4 A per rail)

Size: User	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 1/1